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Generating and testing flying focus laser pulses with Lasy for PIconGPU simulations

— A Bachelors Defense —

Edgar Marquardt

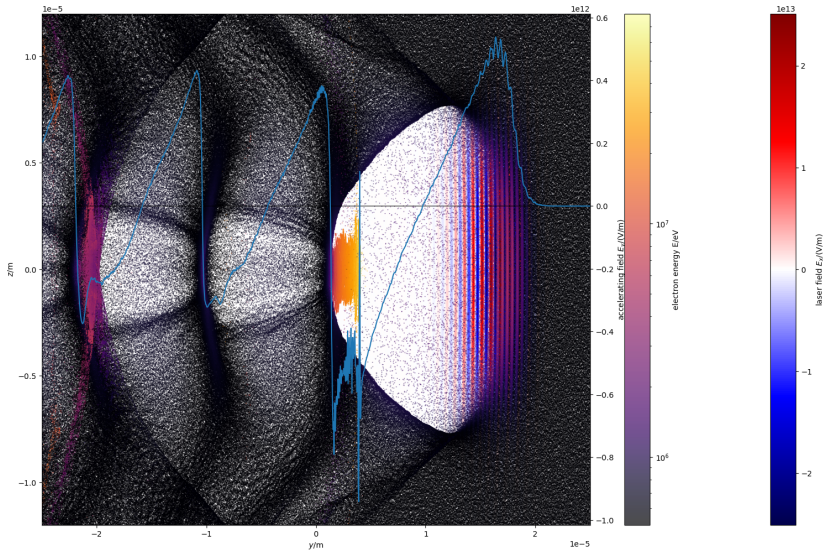
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1 Dephasingless Laser WakeField Acceleration (DLWFA)

Laser WakeField Acceleration (LWFA)



Electric field and electrons in an LWFA (see [1]) simulation.

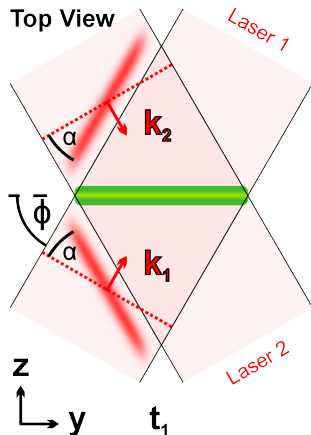
The problem of dephasing

Flying focus lasers – solving the problem of dephasing

1. TWEAC

- **T**raveling-**W**ave **E**lectron **A**Ccelerator
- Developed by Debus et al [2]
- Uses two laser pulses with tilted pulse fronts
- The tilt controls the velocity of the overlapping region

Image: TWEAC setup using two laser pulses. Image taken from Debus [3]

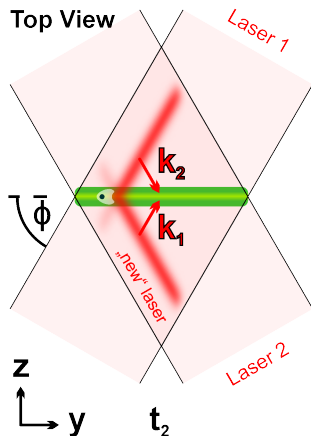


Flying focus lasers – solving the problem of dephasing

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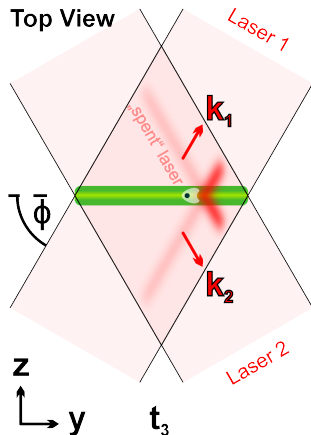


Flying focus lasers – solving the problem of dephasing

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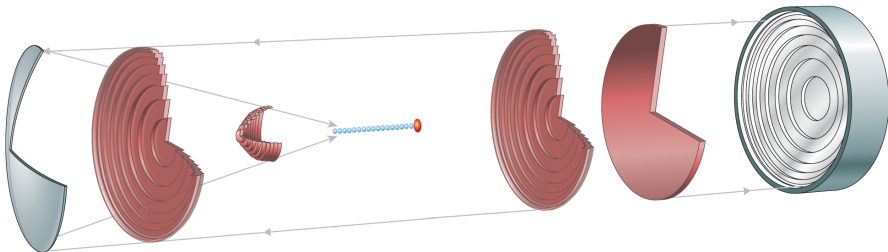
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Flying focus lasers – solving the Problem of dephasing

2. Axiparabola laser

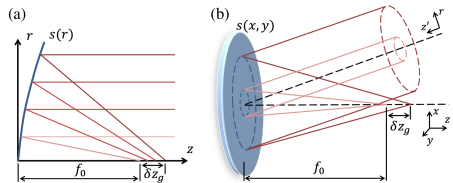


The flying focus setup of Palastro et al [4]. Two optical elements: The Axiparabola (left) and the Radial Group Delay echelon (RGD) (right). Image taken from Palastro et al [4].

Flying focus lasers – solving the Problem of dephasing

2. Axiparabola laser

- Axiparabola [5]:
 - Near-parabolic mirror
 - Focuses light onto a line – the focus region
 - Light at radius r is focused at $f(r) = f_0 + \delta \left(\frac{r}{R} \right)^2$
- Radial Group Delay echelon (RGD) [6][4]:
 - Stepped concentric mirror rings
 - Shape follows some function $\tau_D(r)$
 - controls the timing of the axiparabola focus



Axiparabola functionality. Image taken from Smartsev et al [5].



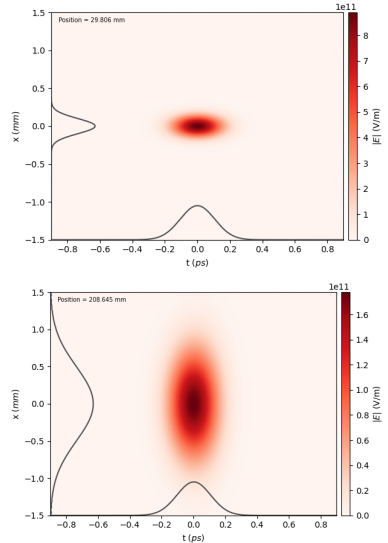
2 Flying focus lasers in Lasy and PIConGPU

Lasy

A python library

- A python library for simulating Laser pulses in a vacuum [7]
- Uses complex envelope of the laser field
- Uses angular spectrum propagation
- Can use cylindrical coordinates for memory and CPU time efficiency
- Offers a range of optical elements

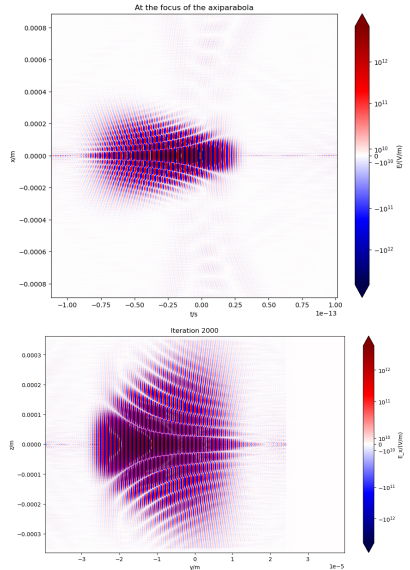
Images: Example of a Gaussian pulse being propagated by Lasy. Top: Generated at the focus. Bottom: $6 z_R$ after the focus.



Importing to PIConGPU

- New module `full_field`
- Generates full electric field and saves it using `openPMD-api`
- `incidentField` method
From `OpenPMDPulse` (see Dietrich [8])
imports the field into PIConGPU

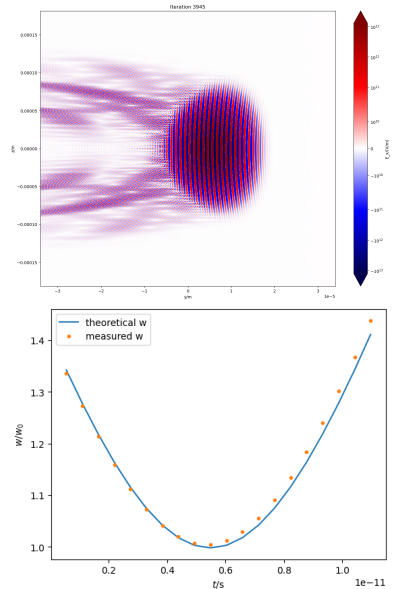
Images: Top: Electric field of the complete flying focus laser at the beginning of the focus region. Bottom: That same electric field entering the simulation window of a PIConGPU simulation.



Testing the method

- Test with Gaussian laser pulse and parabolic mirror
- Some artifacts are visible
 - Problem of the incidentField method
 - See [9]
- Test successful

Images: Top: Electric field of the Gaussian laser pulse z_R before the focus as imported into a PIConGPU simulation.
Bottom: The beam waist w of the pulse over time, compared to theory.

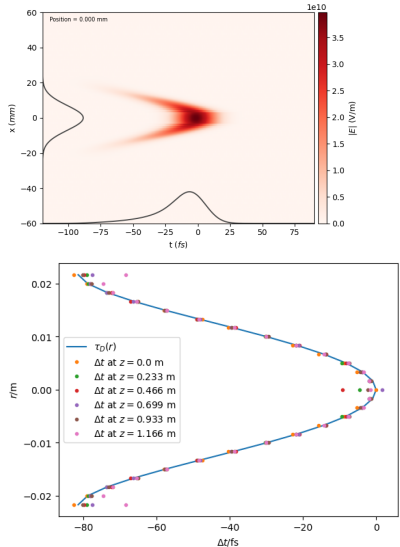


Implementing the flying focus

1. The radial group delay echelon (RGD)

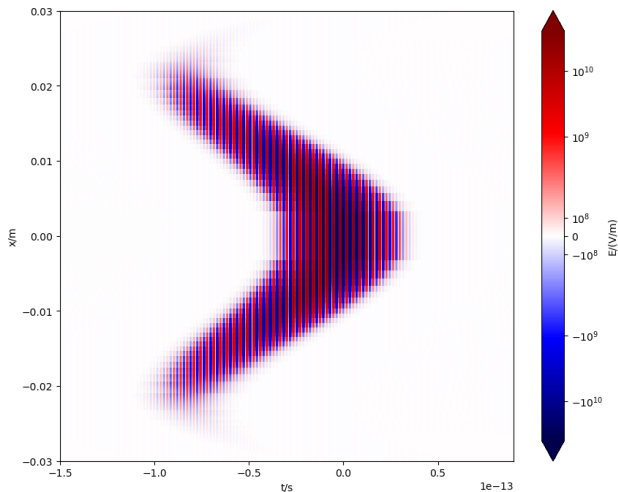
- Implemented from scratch as Lasy optical element
- Following the description by Ambat et al [6]
- Shapes the pulse temporally without focusing or defocusing
- Can generate any radially symmetric shape of delay $\tau_D(r)$

Images: A Gaussian pulse after interacting with the RGD.
Top: Field envelope. Bottom: Test results. Even after long distances the shape still holds.



Implementing the flying focus

1. The radial group delay echelon (RGD)



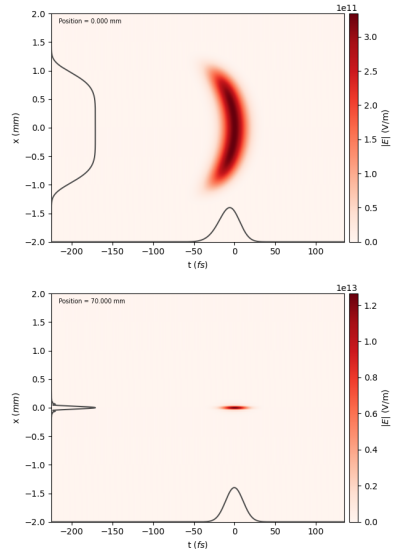
The electric field of the laser after interacting with the RGD.

Implementing the flying focus

2. The axiparabola

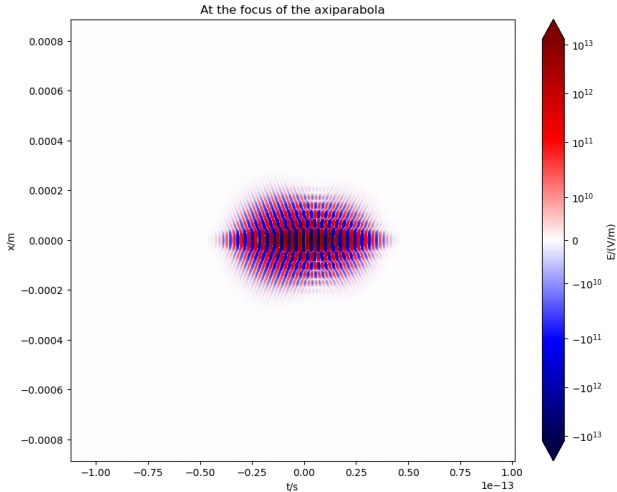
- Included in Lasy
- Following Smartsev et al [5]
- Focusing laser pulse in the focus region
- Also implemented an axiparabola following Ambat et al [6]
 - Very small differences

Images: A super-Gaussian laser pulse after reflecting off the axiparabola. Top: In the near field. Bottom: In the far field at the beginning of the focus region.



Implementing the flying focus

2. The axiparabola



The electric field of the laser at the beginning of the focus region of the axiparabola.

Implementing the flying focus

The complete setup

The simulation had the following steps:

- 1 Initialising the laser pulse with its pulse profile
- 2 Applying the RGD to the laser
- 3 Applying the axiparabola to the laser
- 4 Propagating the laser to the focus region
 - At this point the laser field can be saved to file

Implementing the flying focus

The measurement

- Beam waist w :
 - ?
- Relative focus position $t_{focus} - f_0/c$ or $z - t \cdot c$:
 - ?
- All measurements taken statistically



3 Testing the flying focus lasers

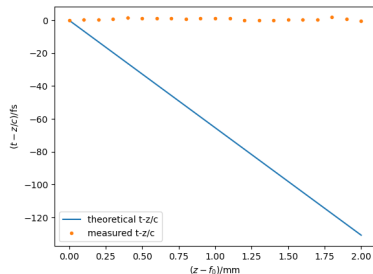
Testing the flying focus laser

First results

- Simulation in Lasy
 - Result: The flying focus effect cannot be seen
- Test failed

Images: Comparing measurements from the simulations to the expected evolution. Top: From a Lasy simulation:

$t - z/c$, Bottom: .

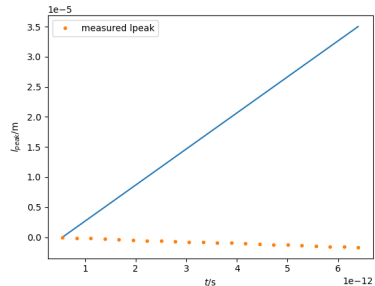


Testing the flying focus laser

The same simulation in PIConGPU

- Simulation in Lasy to the focus region, from there in PIConGPU
- Input file for PIConGPU saved at the beginning of the focus region
- Result: The flying focus effect cannot be seen here either

→ Test failed



Images: Comparing measurements from the simulations to the expected evolution. Top: , Bottom: From a PIConGPU simulation: $z - t \cdot c$.

Testing the flying focus laser

Testing all the used modules



Images:

Testing the flying focus laser

Checking the shapes of the optical elements

- Multiple differential equations for the shape of the RGD $\tau_D(r)$
 - All result in very similar shapes
- Multiple equations for the shape of the axiparabola $s_f(r)$
 - All follow from the same differential equation (and solve it)
 - All are very similar shapes

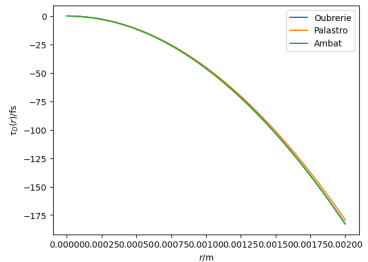


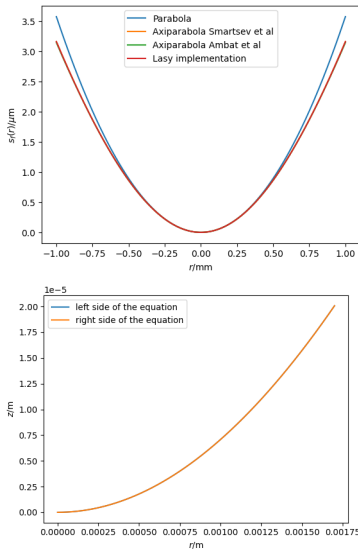
Image: The delay the RGD should achieve according to different sources. The shape of the RGD is similar to this, just stepped in $\lambda_0/2$ steps. Relative differences $\approx 1/50$.

Testing the flying focus laser

Checking the shapes of the optical elements

- Multiple differential equations for the shape of the RGD $\tau_D(r)$
 - All result in very similar shapes
- Multiple equations for the shape of the axiparabola $s_f(r)$
 - All follow from the same differential equation (and solve it)
 - All are very similar shapes

Images: Top: The shape of the axiparabola according to different sources. Relative differences $\approx 10^{-2}$. Bottom: The two sides of the differential equation the axiparabola is derived from. Relative differences $\approx 10^{-5}$.

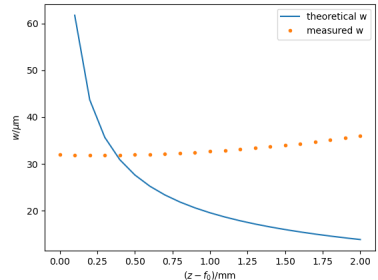
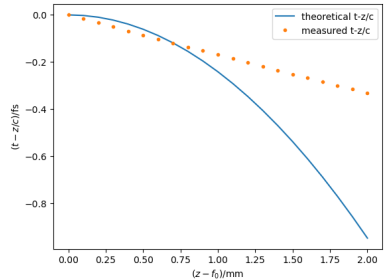


Testing the flying focus laser

Axiparabola only

- With only the axiparabola differences appear already
 - Here must be the problem
- Using the axiparabola described by Ambat et al [6] gives similar results
- Contacted the Lasy developers to no avail (so far)

Images: Comparing measurements from a Lasy simulation with theoretical values using Ambat et al [6]. Top: Arrival time $t - z/c$. Bottom: Beam waist w .





4 Summary and Outlook

- All the code was tested with Gaussian laser pulses
- The shapes of the optical elements were compared to different sources
- the flying focus pulse was simulated and compared in Lasy and PIConGPU
-
- The Lasy developers were contacted

- Easy lasers available in PIConGPU
- LWFA with new laser setups possible
- Some problems still need resolving
- Some tests conducted after writing the bachelors thesis:
 - Trying different propagation methods
 - Initialising the pulse with other pulse profiles
 - Using Cartesian coordinates on nodes with lots of memory

(*) There's more to explore here.

Remaining Possible reasons for failure

- The Axiparabola
 - The shape could still be wrong
- The Propagation
 - Possibly the Lasy propagation gives incorrect/incomplete results
- The Findings in the other papers
 - Maybe the Axiparabola does not work at all
 - Maybe it works in a different manner

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Laser electron-accelerator.
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